

1. Consider the functions $y = x^2$, $f(x) = \frac{1}{4}x^4 + 5x$, $g(t) = (2t + 3)^3 + \pi$.

- a) Compute $\frac{dy}{dx}$.
- b) Compute $\frac{d}{dx}[y]$.
- c) Compute $\frac{dy}{dx}y$.
- d) Compute $\frac{df}{dx}$.
- e) Compute $\frac{d}{dx}[f'(x)] \cdot f(x)$.
- f) Compute $\frac{dg}{dt}$.
- g) Compute $\frac{dg}{dt}g(t)$.
- h) Compute $\frac{dg}{dx}$.
- i) Compute $\frac{dy}{dt}$.

It is important to remember that our derivative “shortcuts” come from the definition of the derivative. This definition is important anytime we want to determine a derivative for which we don’t have a shortcut. It also is important to remember that for a function to be differentiable at a point $x = a$ this means we need $f'(a)$ to exist and this implies that the limit in the definition of the derivative must exist.

- 2. Determine the derivative of $f(x) = 2x - 3$ by using the definition of the derivative and derivative shortcut rules.
- 3. When finding the derivative of $f(x) = 2x - 3$ using shortcut rules you needed to use the *Constant Rule*, the *Constant Multiple Rule*, and the *Difference Rule*. Use the definition to prove these rules.

It is also important not to lose sight of what a *derivative* represents and where they come from. With this in mind, consider the following questions.

4. The function f has the property that $f(4) = 1$ and the tangent line to its graph at $x = 4$ passes through the point $(-2, -3)$.

a) What is $f'(4)$?

b) If $h(x) = 9f(x) - x^2$, find $h'(4)$.

5. Consider the function $f(x) = |x|$ and recall that this function is actually defined as a piecewise function

$$f(x) = |x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}.$$

a) Is $f(x)$ continuous at $x = 0$? Justify your answer.

b) Use the definition of the derivative to determine if $f(x)$ is differentiable at $x = 0$.

c) What does this example tell you about the relationship between continuity and differentiability?